

CO2-AT 2 – SCENARIO BASED ASSESSMENT

Question 1

Scenario 1: Manufacturing Quality Inspection

Problem: During the inspection of metal parts, the captured images contain repeated horizontal stripe patterns caused by electrical noise, making it difficult to identify defects accurately.

1. Type of Noise

- The image is affected by periodic noise, which appears as repeated horizontal stripe patterns caused by electrical interference.

2. Impact on Inspection

- The stripes hide small surface defects such as cracks and scratches.
- Image quality decreases, making defect identification difficult.
- Automated inspection systems may produce incorrect results because of the unwanted patterns.

3. Suitable Enhancement Technique

- Apply Fourier Transform followed by Notch Filtering in the frequency domain.

4. Reason for Choosing This Technique

- Periodic noise is clearly visible as peaks in the frequency spectrum.
- Notch filters remove only the unwanted frequencies while preserving important image details.
- Spatial filtering methods cannot efficiently eliminate periodic stripe noise and may reduce image sharpness.

5. Processing Steps

- Detect periodic interference → Convert image to frequency domain → Apply notch filter → Perform inverse Fourier transform → Obtain a cleaner image for quality inspection.

Question 2

scenario 2: Autonomous Vehicle Navigation

Problem: Images captured by an autonomous vehicle in heavy rainfall are affected by noise, while clear lane markings and road edges must be preserved for accurate navigation.

1. Main Challenge

- Rain introduces unwanted noise into road images while lane markings and road boundaries must stay clear for safe driving.
- Excessive smoothing can remove important edge information needed for navigation.

2. Why Edge Preservation is Important

- Accurate lane detection depends on clear and sharp edges.
- Preserving boundaries helps the vehicle correctly identify road paths.
- Blurred edges may lead to incorrect steering or navigation decisions.

3. Recommended Filtering Method

- Use **Bilateral Filtering**, which smooths noise while maintaining edge details.

4. Why Bilateral Filtering is Effective

- It considers both the distance between pixels and their intensity differences.
- Noise is reduced in flat regions without affecting strong edges.
- It provides better edge preservation than traditional smoothing filters such as Gaussian filtering.

5. Processing Steps

- Capture the noisy road image → Apply bilateral filtering → Reduce rain noise while preserving lane edges → Perform accurate lane detection for safe vehicle movement.